SPORTS ECONOMETRICS & ROSTER OPTIMIZATION BLUEPRINT

Pay-by-Play: Is Pay Following Play?

Testing Compensation Rationality, Production Alignment, and Market Structure in Women's Pro Sports

Project Blueprint
PWHL · WNBA · WPL
AQX Bowl
Specification

OPENING SLIDE HOOK — THE COMPENSATION BENCHMARK

The PWHL's top salary, **Emily Clark's $126,090**, is only **14.8%** of the NHL's 2026–27 minimum. Women's pro leagues are rapidly overhauling how they pay athletes, yet almost no one has rigorously tested whether player compensation tracks on-field production.

1. The Problem, Research Questions & Stakeholders

SECTION 01

Women's pro leagues are changing how they pay players. The **PWHL** just made base salaries public, the **WNBA** has a revenue-linked CBA, and the **WPL** prices players at auction. Almost no one has tested whether that pay tracks production.

Core Research Question: Does pay follow play, and does the pay structure change the answer?

1. How tightly does pay track production in each league?
2. Which players are most under- or overpaid, with statistical confidence?
3. Which structure tracks production best: a salary-band system, a tiered cap, or an auction?

Who Uses It

- **General Managers (GMs):** A quantitative bargain finder and an integer-programming roster optimizer.
- **Players & Agents:** Objective empirical evidence for contract negotiations. As the PWHLPA president noted, salary disclosure helps players negotiate better contracts.

2. Data Sources, Verified Facts & Caveats (Part I: North America)

SECTION 02 · PWHL & WNBA

LEAGUE	PAY DATA & CAP RULES	PERFORMANCE LOADERS	VERIFIED CAVEATS & DATA-QUALITY TRAPS
PWHL Ice Hockey	Union salary guide lists **base salaries only** (no bonuses, incentives, benefits, or housing). Accurate as of April 12; updated annually. **Pay Rule:** Team average salary must stay within 10% of the **$60,099.99** target in 2026–27.	`pwhl-scraper` pulls from the HockeyTech API. `fastRhockey` (R) provides PWHL box score and play-by-play loaders.	• One outlet counts **194 players**; another says **197**. • **15 players** signed mid-season aren't listed. • Outlets disagree on minimum-salary count (**16 vs. 17**). • **Fix:** Reconcile against published team totals as checksums (e.g., Boston's 25 contracts totaling **$1,436,173.50**).
WNBA Basketball	**Spotrac:** One tracker follows 133 signings verified via Spotrac. **2026 Context:** $7M cap; minimum-salary players got raises of **300%+**; **30+ players** earn $1M+ (vs. none	`wehoop` provides `load_wnba_player_box` and player-impact loaders.	• Spotrac isn't an official source; outlets disagree on the top tier (**$1.4M vs. $1.67M**)—use one source throughout. • One version of `wehoop` docs shows a **404 error** for the 2026 play-

2. Data Sources, Verified Facts & Caveats (Part II: WPL & NWSL)

SECTION 02 · CONTINUED

WPL (Women's Cricket) — Auction Pricing

- **Pay:** Auction price plus published retention prices. The 2026 auction sold **67 players for over ₹40 crore**. Because no clean download is confirmed, compile prices into a CSV by hand.
- **Performance:** Cricsheet's open ball-by-ball data covers **88 WPL matches**. Because this is a small sample, empirical Bayes shrinkage is essential.
- **Timing:** The 2026 auction had **277 registered players for 73 slots**, and the season started **January 7**—enabling direct comparison of pre-season prices against in-season performance.

NWSL: The "Transparency Gap" Slide

Only a handful of top NWSL salaries have been publicly reported, meaning the league **cannot be tested empirically**. In the presentation, NWSL serves as the dedicated transparency gap case: demonstrating why union salary disclosure (like the PWHL's) is essential for market accountability.

3. Unified Data Model

SECTION 03 · SCHEMA

All three leagues map into a standardized player-season table so downstream shrinkage, percentile ranking, and dashboard modules run on an identical schema:

FIELD GROUP	STANDARDIZED COLUMNS	ROLE IN ANALYTICAL PIPELINE
Identity & Context	`league`, `season`, `player_id`, `name`, `team`, `position`, `age`	Stratifies players within league and position cohorts (`forward` / `defender` / `goalie`, court role, or cricket phase).
Compensation	`pay_raw`, `pay_type (salary|auction)`, `pay_pct`	Stores local currency amount alongside within-league, within-position percentile rank (`pay_pct`).
Exposure & Raw Value	`games`, `exposure (minutes | balls faced/bowled)`, `value_raw`	Measures sample size (n) for shrinkage weighting and raw sport-specific production composite.
Shrunk Value & Gaps	`value_shrunk`, `ci_low`, `ci_high`, `gap`, `rookie_flag`, `draft_pick`, `data_flags`	Stores empirical Bayes estimate, 80% bootstrap bounds, percentile surplus (`gap`), entry status, and QA flags.

4. Method: Sport Value Scores & Pay Comparability

SECTION 04 · 4.1 – 4.2

- **4.1 Value Score — PWHL Skaters & Goalies:** Goals, assists, shots, and penalties drawn minus taken, per game, within position (forward or defender). Add on-ice goal share if play-by-play allows. Model goalies separately.
- **4.1 Value Score — WNBA:** A per-minute box-score composite scaled by minutes, with `wehoop`'s player-impact loader as a cross-check.
- **4.1 Value Score — WPL:** Runs added or saved against phase baselines (powerplay, middle, death) from ball-by-ball data, with each wicket valued in runs by phase.
- **Replacement Level Benchmark:** Defined uniformly as the **20th percentile of regulars** at that

4. Method: Shrinkage, Two-Lens Test & Roster Optimization

SECTION 04 · 4.3 – 4.6

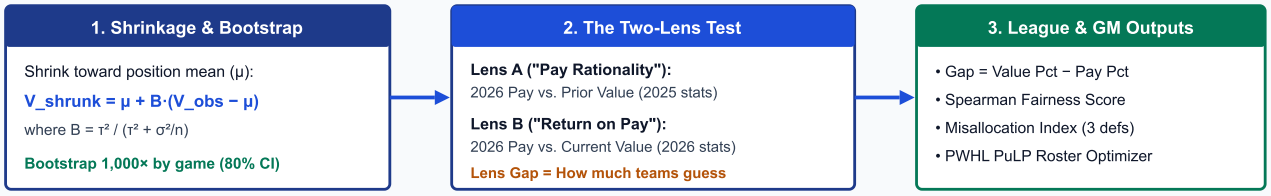


Figure 1: Analytical pipeline linking empirical Bayes shrinkage, game-level bootstrap intervals, and the two-lens evaluation framework.

4.3 Gap, Shrinkage & Intervals

- **Percentile Gap:** $Gap = Value\ Percentile - Pay\ Percentile$. A positive gap means underpaid.
- **Model-Based Version:** Fit mixed-effects regression $log(pay) \sim value + position + age + rookie + (1 \mid team)$.
- **Shrinkage toward Position Mean:** $V_{shrunk} = \mu + B \cdot (V_{obs} - \mu)$, where $B = \tau^2 / (\tau^2 + \sigma^2 / n)$.
- **Game Bootstrap:** Resample by game **1,000 times**. Label a player *"confidently underpaid"* only if the **80% interval** stays strictly above zero.

4.4 Two Lenses

- **Lens A ("Pay Rationality"):** Pay vs. prior value (e.g., 2026 WNBA salary against 2025 stats).
- **Lens B ("Return on Pay"):** Pay vs. current value (e.g., the same salary against 2026 stats).
- **Interpretation:** The gap between the two lenses measures how much teams are guessing.

4.5 Cross-League Scores

- **Fairness Score:** The Spearman rank correlation between pay and value, reported with bootstrap intervals.
- **Misallocation Index:** The share of total payroll that would have to move to make pay match value share exactly. Easy to explain, but sensitive to value definitions—report under **three alternative value definitions**.

4.6 Roster Optimizer (PWHL Only)

Formulate an integer linear program (`PuLP`) to maximize total shrunk value subject to: (1) position minimums, (2) observed roster size in the data, and (3) the ±10% average-salary band around $60,099.99. Outputs a **"best roster for the budget"** and specific player swaps.

5. Seven-Point Validation Plan

SECTION 05 · QA & RIGOR

1. **Checksum Reconciliation:** Reconcile the PWHL dataset against published team totals (e.g., Boston's $1,436,173.50).
2. **Split-Half Reliability:** Compute odd-game vs. even-game correlation for each sport's value metric.
3. **Year-over-Year Persistence:** Test season-to-season stability of player value scores.
4. **Held-Out Coverage:** Check empirical coverage of the 80% bootstrap intervals on held-out games.
5. **Metric Robustness:** Verify Misallocation Index and rankings across three alternative value definitions.
6. **Placebo Test:** Shuffle pay randomly within position and verify that the pay-value correlation collapses to zero.
7. **Optional Live Pre-Commitment:** The next WPL auction is **October 28 in Kochi**. If deadlines permit, commit price predictions to GitHub before bidding starts and score them post-auction (skip if WPL model isn't ready).

6. The Interactive Application (Streamlit + Plotly)

SECTION 06 · PRODUCT ARCHITECTURE

The decision-support web application translates the econometric models into six interactive views tailored for GMs, players, and agents:

1. **Scoreboard:** High-level cross-league comparison displaying the Spearman fairness score, payroll misallocation %, and the count of confident bargains for each league.
2. **League Explorer:** Interactive pay-percentile vs. value-percentile Plotly scatter plot with clearly labeled performance-compensation quadrants.
3. **Raise Radar:** Ranked board of the top underpaid players featuring 80% bootstrap intervals and an auto-generated one-line negotiation case for each athlete.
4. **Player Card:** Individual multi-game performance trend, contract context, and nearest statistical comparables earning *"similar pay, more value"* (or vice versa).
5. **Roster Lab:** Interactive PWHL integer linear programming (`PuLP`) roster builder displaying the optimal roster for the salary band and recommended trade/signing swaps.
6. **Methods & Limits:** A transparent documentation tab explaining shrinkage, bootstrap intervals, and what box-score and play-by-play numbers cannot capture.

7. Hypotheses to Test (Empirical Questions, Not Pre-Set Results)

SECTION 07 · RESEARCH HYPOTHESES

Rather than assuming conclusions in advance, the analytical pipeline is structured to test four falsifiable hypotheses across the three leagues:

HYPOTHESIS H1 · DRAFT PEDIGREE VS. PLAY

Does draft slot predict pay better than production? In the PWHL, the No. 1 overall pick earns **$90,500** while the No. 2 pick earns **$60,000**. We test how long entry draft pedigree dominates actual on-ice value share.

HYPOTHESIS H2 · CBA STRUCTURAL SHIFT

Mechanical vs. performance raises in the WNBA: With the 2026 cap jumping to $7M and minimums rising 300%+, how much of the 2025-to-2026 raise was purely mechanical across tiers, and how much followed prior-season performance?

HYPOTHESIS H3 · AUCTION REPUTATION BIAS

Do auctions price reputation or output? In the WPL auction, star wicketkeeper Alyssa Healy went unsold while all-rounder Amelia Kerr commanded **₹3 crore**. We test whether bidding tracks recent phase-adjusted output or career fame.

HYPOTHESIS H4 · INSTITUTIONAL DESIGN

Which mechanism aligns pay and play best? Does a compressed team-average band (**PWHL**) track production more or less tightly than a tiered salary cap (**WNBA**) or a live player auction (**WPL**)?

Analytical Framing Rule: "Who Deserves a Raise"

Across all application views and hypothesis tests, frame player-level findings positively around **"who deserves a raise"** (identifying undervalued contributors and confident bargains via Raise Radar) rather than punitive "who is overpaid" leaderboards. A statistical gap is a starting point for negotiation and roster construction, not proof of front-office error.

8. Mapping to the Judging Criteria

SECTION 08 · EVALUATION

JUDGING CRITERION	WHAT YOU SHOW IN PAY-BY-PLAY
Practical Application	**Raise Radar** with auto-written negotiation briefs for players and agents; **Roster Lab** (`PuLP` optimizer) and confident bargain boards for General Managers.
Analytical Insight	Empirical Bayes shrinkage (V_{shrunk}), 1,000× game-level bootstrap intervals, the Two-Lens test (rationality vs. return), placebo shuffling, and 3-definition robustness checks.
Data Presentation	One unified cross-league scoreboard, an interactive quadrant scatter plot, and plain-English verdicts.

9. Repository Architecture & Execution Build Order

SECTION 09 · ENGINEERING PLAN

```
pay-by-play/ ├── data/{raw,processed} ├──
src/ingest_{pwhl,wnba,wpl}.py ├──
src/value/{pwhl,wnba,wpl}.py ├──
src/model/{normalize,gap,shrink,twolens}.py ├──
src/optimize/pwhl_roster.py ├──
app/streamlit_app.py ├──
notebooks/validation.ipynb └── README.md
(method, limits, how to run)
```

Sequential Build Order

1. **PWHL end to end** (ingest, checksums, value, shrinkage, `PuLP` optimizer, Streamlit skeleton).
2. **WNBA** (Spotrac + `wehoop` verification & Two-Lens pipeline).
3. **WPL** (manual auction CSV + Cricsheet phase value).
4. **Cross-league page, validation suite, and final pitch.**

Scope Rule: If time runs short, ship two leagues well rather than three thinly.

10. Risks & Guardrails

SECTION 10

- **Unobserved Value:** A gap isn't proof of a mistake. Pay can reflect drawing power, national-team status, or leadership stats miss. State this clearly on screen.
- **Terms of Use:** Scraping terms for Spotrac and news auction lists couldn't be confirmed, and the PWHL salary page requires manual verification.
- **Scope Protection:** Three leagues is ambitious—protect validation time.
- **Tone:** Frame as *"who deserves a raise,"* not *"who's overpaid."*
- **Originality:** Verify no past AQX Bowl project overlaps.

11. Pitch Outline (3 Minutes)

SECTION 11

1. **The Hook:** Emily Clark's $126,090 top PWHL salary vs. NHL minimum (14.8%).
2. **The Question:** "Does pay follow play, and does structure matter?"
3. **The Scoreboard:** Cross-league view + one surprising league comparison.
4. **One Player Story:** A confident bargain highlighted with her 80% interval.
5. **Roster Lab Demo:** Live PWHL budget-constrained optimization swaps.
6. **The NWSL Gap:** Salary transparency is what makes accountability possible.
7. **Limits & Ask:** Honest caveats and a clear call to action for GMs and players.